

ANTHONY TY MARLER

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Senior Application Architect | Inventor | Energy and Climate-Tech | AI-Assisted Systems | Fractional CTO

SUMMARY

Senior application architect and fractional CTO currently shipping AI-assisted products with Cursor, Claude, and ChatGPT. Previously spent five years at Vistra designing generation, mining, and emissions systems, including a cloud credit-trading platform that accounted for \$60 million in profit in the first two months. Named inventor on two U.S. patents. Combines AWS, .NET, SQL, and systems integration with energy, mining, and manufacturing domain expertise, and leads cross-functional teams through the full project lifecycle.

SKILLS

Application Architecture, Generative AI, Cursor, Claude, ChatGPT, Fractional CTO, Systems Integration, Amazon Web Services (AWS), Chef, Microsoft .NET, C#, VB.NET, C++, SQL, PHP, Java, Agile, Scrum, Cloud Computing, Data Modeling, Database Design, Root-Cause Analysis, TCO Analysis, Cross-Functional Leadership, Tableau, SharePoint, OSIsoft PI, GIS, ESRI GIS, SCADA, PLC, Amazon RDS, MS SQL Server, MySQL, Oracle, DB2, APIs, StackVision, NetDAHS, Formotus, TDMobile, Visual Studio, Emissions Compliance, Carbon Credit Trading, Mining Operations Software, MSHA Training, Biofuels Process Intensification

EXPERIENCE

Application Architect, AI-Assisted Systems

Independent, Texas

March 2025 - Present

- Used Cursor, Claude, and ChatGPT to architect and develop an international fugitive-tracking application and a custom architectural-planning website.
- Delivered a clothing-brand retail / e-commerce site and an internet television streaming application with AI-assisted architecture, UI, and backend development.

Senior Analyst / Architect, Generation

Vistra Corporate Services Company, Texas

December 2019 - February 2025

- Designed and developed a data-driven process engine to manage dynamic tasks, providing automated steps to drive the process to completion.
- Designed and developed mine planning and monitoring software to calculate productivity and measure results by shift.
- Designed and developed a cloud-based AWS platform that ingested SCADA, PLC, and third-party emissions data into a normalized database for credit trading; accounted for \$60 million in profit in the first two months, cut trading latency from weeks to hours, and reduced compliance reporting time by 80%.
- Led web applications supporting mining, fossil generation, battery, and solar operations, including Tableau dashboards for traders and compliance officers.

Business Application Manager

Accenture, Texas

August 2016 - December 2019

- Led day-to-day maintenance, enhancement, and operations for the Fossil Applications portfolio serving Luminant generation and mining.
- Directed development across AWS, Chef, .NET, C#, C++, PHP, SQL Server, MySQL, Tableau, Formotus, TDMobile, StackVision, and NetDAHS.
- Owned troubleshooting and root-cause analysis for production issues across ROTT, Skaief, SmartProcedures, GIS, Mobility, Safety Index, Proact, Digital Signage, Qdabra, SharePoint, PI, and land management.
- Led the Fossil Applications cloud modernization on AWS and Chef, putting DevOps, QA, and engineering on one release path and finishing the cutover three weeks early.

Senior Application Architect

Capgemini, LLC, Texas

February 2008 - August 2016

- Architected and delivered Fossil Generation applications covering MSHA training, emissions compliance, procedures, GIS, safety, mobility, digital signage, and SharePoint.
- Provided architectural solutions for new builds and rebuilds to improve reliability and functionality, including TCO analysis of existing and future-state environments.
- Served as SME for mining operations, mining software systems, data modeling, and database design; introduced mobility (Formotus, TDMobile), OSIsoft PI, land management, and application monitoring.

Owner

Texas Built Company, LLC, Winnsboro, TX

February 1998 - February 2008

- Invented a process-intensified biodiesel reactor that reduced reaction time from four hours to four minutes, enabling continuous rather than batch production, about 60x throughput in the same footprint, and roughly 40% lower energy cost per gallon while maintaining ASTM fuel quality.
- Patent pending: Texas Built process intensified biofuels reactor technologies.

Principal Engineer / Worldwide Software Systems Architect

Samsung / AST Research, United States

March 1997 - February 1998

- Worldwide software systems architecture for PC manufacturing and configuration platforms.

Senior Systems Engineer

Compaq Computer Corporation, Houston, TX

October 1993 - March 1997

- Invented and developed a patented software-delivery system that eliminated more than 70% of the manufacture time of Compaq computers. Also invented a post-sales configuration method that increased inventory flexibility.
- Named inventor on U.S. Patents 6,202,070 and 5,974,567.

Owner

Automated Response Information Systems, Inc., United States

October 1991 - October 1993

- Designed and implemented a hardware-based voice-recognition system for hands-free data collection in harsh environments (early applied AI / speech recognition).

Systems Engineer

Stanford Telecommunications, Inc., United States

July 1989 - August 1991

- Worked on a team to develop planning and control software for the defense communication satellite network.

Field Engineer

Elcom and Associates, United States

February 1987 - July 1989

- Field engineering and technical support for communications and electronics systems.

Strategic Microwave System Repair Technician

U.S. Army, United States

February 1984 - February 1987

- Maintained strategic microwave communications systems. Completed U.S. Army Basic Electronics, Communications Electronics Strategic Microwave System Repair, and Cryptographic courses.

EDUCATION

Colorado Technical College

Computer Science coursework and Certificate in Ada

Compaq Computer Corporation

Advanced C++ and Advanced Java

Professional Development

DEC VAX/VMS System Management

U.S. Army

Basic Electronics; Communications Electronics Strategic Microwave System Repair; Cryptographic courses

PATENTS

- Pending - Texas Built process intensified biofuels reactor technologies.
- U.S. 6,202,070 (issued March 13, 2001) - Computer manufacturing system architecture with enhanced software distribution functions (Compaq Computer Corporation; named inventor).
- U.S. 5,974,567 (issued October 26, 1999) - Ghost partition / runtime-selectable file system (Compaq Computer Corporation; named inventor).